Xi'an Jiaotong-Liverpool University 西交利物浦大学

PAPER CODE	EXAMINER	ACADEMIC UNIT	TEL
MTH004		Foundational Mathematics	

2nd SEMESTER 2023-24 FINAL EXAMINATION

Time allowed: 120 minutes

INSTRUCTIONS TO CANDIDATES

1. This is a **closed-book** examination, which is to be written without books or notes.
2. The total mark is 100 and there are 22 questions in total.
3. **Calculators are not allowed**.
4. This exam consists of two sections:
   Section I consists of 18 multiple choice questions worth 3 marks each for a total of 54 marks. There is only one correct answer for each question and answers should be filled into MCQ card.
   Section II consists of 4 comprehensive problems and worth 46 marks in total. The process of solution should be provided and solutions should be written in the Answer Booklet, otherwise 0 marks will be given.
5. Answer all questions. There is NO penalty for providing a wrong answer.
6. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Do not open this question booklet until instructed to do so!

I. Choose the correct answer. (3 marks for each question, 54 marks in total)

MCQ

II. Calculations and comprehensive problems. (46 marks in total)

19.(12 marks) Let $f(x, y) = xe^{y^2}$.

(1) Find the gradient ∇f.

(2) Determine in which direction f INCREASES most quickly at $(1, 1)$. And at what rate?

(3) Using total differential to estimate $f(1.1, 1.1)$.

(4) Find $\left[\frac{d}{du} f(\sqrt{u}, u^2)\right]_{u=1}$.

20.(12 marks) Find the maximum and minimum value of $f(x, y) = e^{xy}$ on $D = \{(x, y)|x^2 + 4y^2 \leq 2\}$.

21.(12 marks)

(1) Evaluate the volume of the solid bounded by $S_1 : z = 20 - 4x^2 - y^2$ and $S_2 : z = x^2 + 4y^2$.

(2) Given $\Pi : 8x + 4y + z = 28$, find the area of Π in the first octant.

22.(10 marks) Find $g(x)$ such that $g(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^{2n+1}}{2n(2n-1)}$.

End of Exam Paper

Xi'an Jiaotong-Liverpool University 西交利物浦大学

PAPER CODE	EXAMINER	ACADEMIC UNIT	TEL
MTH004		Foundational Mathematics	

2nd SEMESTER 2022/23 FINAL EXAMINATION MULTIVARIABLE CALCULUS (BUSINESS AND ARTS)

TIME ALLOWED: 120 MINUTES

INSTRUCTIONS TO CANDIDATES

1. This is a **closed-book** examination, which is to be written without books or notes.
2. The total mark is 100 and there are 22 questions in total.
3. **Calculators are not allowed**.
4. This exam consists of two sections:
   Section I consists of 18 multiple choice questions worth 3 marks each for a total of 54 marks. There is only one correct answer for each question and answers should be filled into MCQ part in Answer Sheet.
   Section II consists of 4 comprehensive problems and worth 46 marks in total. The process of solution should be provided and solutions should be written in the blank for each question in Answer Sheet, otherwise 0 marks will be given.
5. Answer all questions. There is NO penalty for providing a wrong answer.
6. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Do not open this question booklet until instructed to do so

I. Multiple Choice Questions. (3 marks for each question, 54 marks in total)

II. Comprehensive problems. (46 marks in total)

19. (12 marks) Given the surface $z = f(x, y) = \sqrt{2 + x^2 - 2y^2}$, and the point $Q(1, 1)$.

(1) Find the partial derivatives $\left.\frac{\partial z}{\partial x}\right|_{(1,1)}$ and $\left.\frac{\partial z}{\partial y}\right|_{(1,1)}$.

(2) Let C be the curve intersecting the surface $z = f(x, y)$ and the plane $x = 1$. Find the parametric equations of the line tangent to C at $P(1, 1, 1)$.

(3) Find the directional derivative of $f(x, y)$ at Q in the direction $\mathbf{u} = 2\mathbf{i} - \mathbf{j}$.

(4) In what direction does f increases most rapidly at Q ? What is the rate of change of f at the point Q in that direction?

20. (12 marks)

Assuming that the marketing cost of online advertising is x (in million of yuan) and the marketing cost of television advertising is y (in million of yuan), the company's revenue R is a function of x and y:

$$R(x, y) = 20 + 10x + 6y - xy - 2x^2 - y^2.$$

(1) Find the local extreme value of $R(x, y)$.

(2) If the budget for the total marketing costs is 2 (in million of yuan), how should this amount be allocated in order to maximize the revenue R?

21. (12 marks)

(1) Evaluate the double integral $\iint\limits_S \sqrt{x^2 + xy}\ \ dA$, where S is the region bounded by the lines $y = x, x = 1, x = 2$ and $y = 0$.

(2) Find the volume of the solid bounded by the surfaces $z = 12 - 2x^2 - y^2$ and $z = x^2 + 2y^2$.

22. (10 marks) Consider the power series $\sum_{n=0}^{\infty}(n + 2)x^n$.

(1) Find the convergence set for this power series.

(2) Find the sum of this power series.

End of Exam Paper

2021-22 2nd SEMESTER FINAL EXAMINATION (Year 1)
MUTIVARIABLE CALCULUS (BUSINESS AND ARTS)

Time allowed: 120 minutes
Total marks: 100; Total questions: 24.

II. Calculations and comprehensive problems. (40 marks in total)

21. (12 marks)

(1) Let $z = \ln(x-y), x = se^t, y = e^{st}$. Find $\dfrac{\partial z}{\partial t}$ by the chain rule.

(2) Let $z = z(x,y)$ be an implicit function defined by the equation $x^3 + 3y^2z - xyz^3 = 0$. Find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$.

22. (8 marks)

Consider the Cobb-Douglas production model for a manufacturing process: $P(x,y) = kx^{\frac{1}{3}}y^{\frac{2}{3}}, (k > 0)$, where x and y are two inputs with unit costs 3 and 2 respectively. If the cost constraint is $3x + 2y = 54$, determine x and y to maximize the production.

23. (8 marks)

Find the volume of the solid bounded by $z = 2\sqrt{x^2+y^2}$ and $z = 3 - (x^2+y^2)$.

24. (12 marks)

Let $I_n = \iint\limits_S \dfrac{(\arcsin y)^{n-1}}{\sqrt{(1-x^2)(1-y^2)}} dA$ $(n = 1, 2, \cdots)$, where S is the plane region bounded by the x-axis, $y = x$ and $x = \dfrac{1}{2}$.

(1) Find I_n.

(2) Let $a_n = (n+1)I_n$. Find the convergence set and the sum of the power series $\sum\limits_{n=1}^{\infty} a_n x^n$.

End of Exam Paper

Appendix

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots \qquad -1 < x \leq 1$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots \qquad -1 \leq x \leq 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots$$

Xi'an Jiaotong-Liverpool University 西交利物浦大学

PAPER CODE	EXAMINER	ACADEMIC UNIT	TEL
MTH004		Foundational Mathematics	

2nd SEMESTER 2023-24 RESIT EXAMINATION

Time allowed: 120 minutes

INSTRUCTIONS TO CANDIDATES

1. This is a **closed-book** examination, which is to be written without books or notes.
2. The total mark is 100 and there are 22 questions in total.
3. **Calculators are not allowed**.
4. This exam consists of two sections:
   Section I consists of 18 multiple choice questions worth 3 marks each for a total of 54 marks. There is only one correct answer for each question and answers should be filled into MCQ card.
   Section II consists of 4 comprehensive problems and worth 46 marks in total. The process of solution should be provided and solutions should be written in the Answer Booklet, otherwise 0 marks will be given.
5. Answer all questions. There is NO penalty for providing a wrong answer.
6. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Do not open this question booklet until instructed to do so!

I. Choose the correct answer. (3 marks for each question, 54 marks in total)

MCQ

II. Calculations and comprehensive problems. (46 marks in total)

19.(12 marks) Consider upper half of the two-sheet hyperboloid S described by function $z = f(x, y) = \sqrt{x^2 + 3y^2 + 3}$ and a point $P(1, 2, 4)$ on the surface.

(1) Calculate $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at P.

(2) Find the equation of tangent plane Π of S at point P.

(3) Find the parametric equations of the intersection line of Π and the plane $y = 2$.

20.(12 marks)

(1) Given three non-zero numbers a, b, c. Show that the equation of the plane which intersect coordinate axis with $(a, 0, 0)$, $(0, b, 0)$, $(0, 0, c)$ is $\dfrac{x}{a} + \dfrac{y}{b} + \dfrac{z}{c} = 1$.

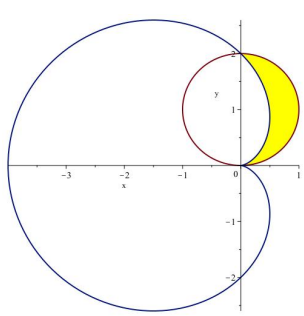
(2) Assume that $a > 0$, $b > 0$ and $c > 0$. Find the value of a, b, c such that the plane passes through point $(3, 2, 1)$ and cuts off the smallest volume in the first octant. (Hint: Volume of a tetrahedron can be computed by $V = \dfrac{1}{3} \cdot (\text{area of base}) \cdot (\text{height})$.)

21.(12 marks) Let $\gamma_1 : x^2 + (y-1)^2 = 1$ and $\gamma_2 : x^2 + y^2 - 2\sqrt{x^2 + y^2} + 2x = 0$ be curves on the xy-plane. Let D be the region that outside γ_2 but inside γ_1.

(1) Rewrite γ_1 and γ_2 in polar coordinate.

(2) Find the area of D (colored in yellow below).

(3) Find the volume between $f(x, y) = \begin{cases} \frac{x+y}{x^2+y^2}, & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$ and the xy-plane over D.



22.(10 marks) Does $\lim_{x\to 0} \dfrac{\arctan(x^3) + x\cos x - \sum_{n=1}^{\infty} \frac{x^{2n-1}}{n}}{x^5}$ exist? Calculate the limit if so or state why the limit does not exist.

End of Exam Paper

Xi'an Jiaotong-Liverpool University 西交利物浦大学

PAPER CODE	EXAMINER	ACADEMIC UNIT	TEL
MTH004		Foundational Mathematics	

2nd SEMESTER 2022/23 RESIT EXAMINATION MULTIVARIABLE CALCULUS (BUSINESS AND ARTS)

TIME ALLOWED: 120 MINUTES

INSTRUCTIONS TO CANDIDATES

1. This is a **closed-book** examination, which is to be written without books or notes.
2. The total mark is 100 and there are 22 questions in total.
3. **Calculators are not allowed**.
4. This exam consists of two sections:
   Section I consists of 18 multiple choice questions worth 3 marks each for a total of 54 marks. There is only one correct answer for each question and answers should be filled into MCQ part in Answer Sheet.
   Section II consists of 4 comprehensive problems and worth 46 marks in total. The process of solution should be provided and solutions should be written in the blank for each question in Answer Sheet, otherwise 0 marks will be given.
5. Answer all questions. There is NO penalty for providing a wrong answer.
6. All materials must be returned to the exam invigilator upon completion of the exam. Failure to do so will be deemed academic misconduct and will be dealt with accordingly.

Do not open this question booklet until instructed to do so

I. Multiple Choice Questions. (3 marks for each question, 54 marks in total)

II. Comprehensive problems. (46 marks in total)

19. (12 marks) Let $f(x, y, z) = x^3 e^{y^2 z}, P(-1, 1, 1)$.

(1) Find the gradient of $f(x, y, z)$ at P.

(2) Find the directional derivative of $f(x, y, z)$ at P in the direction $\mathbf{u} = 2\mathbf{i} - \mathbf{j} + 2\mathbf{k}$.

(3) In what direction does f increase most rapidly at P ? What is the rate of change of f at the point P in that direction?

(4) In what direction the directional derivative of $f(x, y, z)$ at P is zero?

20. (12 marks)

Find the global maximum value and global minimum value of $z = f(x, y) = x^2 + 3y^2 - 4x$ on the set $S = \{(x, y)|(x-1)^2 + y^2 \leq 9\}$.

21. (12 marks)

(1) Evaluate the double integral $\iint\limits_S (x+y)dA$, where S is the region enclosed by the triangle with vertices $(0, 0), (1, 4)$ and $(2, 4)$.

(2) Find the volume of the solid bounded by the surface $z = e^{x^2+y^2}\left(x^2 + y^2\right)$, the plane $z = 0$ and the cylinder $x^2 + y^2 = 1$.

22. (10 marks)

(1) Find the Maclaurin series representation of $f(x) = \ln(2 + 3x - 2x^2)$.

(2) Find the convergence radius and convergence set for this Maclaurin series.

End of Exam Paper

2021-22 2nd SEMESTER RESIT EXAMINATION (Year 1)
MUTIVARIABLE CALCULUS (BUSINESS AND ARTS)

Time allowed: 120 minutes
Total marks: 100; Total questions: 22.

II. Calculations and comprehensive problems. (46 marks in total)

19. (12 marks)

(1) Let $z = u^2 \ln v, u = \dfrac{x}{y}, v = 3x - 2y$. Find $\dfrac{\partial z}{\partial y}$ by the chain rule.

(2) Let $z = z(x, y)$ be an implicit function defined by the equation $e^z - xyz = 0$. Find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$.

20. (14 marks)

(1) Find the local maximum and minimum values, if any, of $f(x, y) = x^3 + y^3 + 3xy$.

(2) Let Ω be a rectangular box without a lid. Find the dimensions of Ω with surface area $108ft^2$ that has maximum volume and find the maximum volume.

21. (12 marks)

(1) Find the double integral $\iint\limits_S xydA$, where S is the region bounded by $x^2 + y^2 = 2y, y = x$ and y-axis.

(2) Find the volume of the solid in the first octant bounded by the coordinate planes and the planes $2x + y - 4 = 0$ and $8x + y - 4z = 0$.

22. (8 marks)

Let S be the region bounded by $y = x, x^2 + y^2 = x$ and x-axis.

$$a_n = \iint\limits_S \frac{y^n}{(x^2 + y^2)^{\frac{n+1}{2}}} dA.$$

Find $\sum\limits_{n=0}^{\infty} a_n$.

End of Exam Paper

Appendix

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots \qquad -1 < x \leq 1$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots \qquad -1 \leq x \leq 1$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots$$